HEAT TREATMENTS

- Definition:

  \- ***a combination of heating and cooling operations timed and applied to a metal or alloy in the solid state to produce desired properties***

Objectves/ Purposes of heat treatment

- To improve mechanical properties
- To improve surface hardness
- To increase resistance to wear, corrosion etc
- To improve machinability
- To relieve internal stresses
- To refine grain structure after hot working
- To improve magnetic and electrical proprties

Stages of HT

- Heating to required temperature
- Holding the material at this temperature for a specified period of time (soaking)
- Cooling the steel at specified rate.

HEAT TREATMENT PROCESSES

- Annealing – to soften the steel
- Normalising – to refine the structure
- Hardening – to increase the hardness
- Tempering – to eliminate brittleness in hardened steel
- Surface hardening – to provide wear resistance surface with tough core

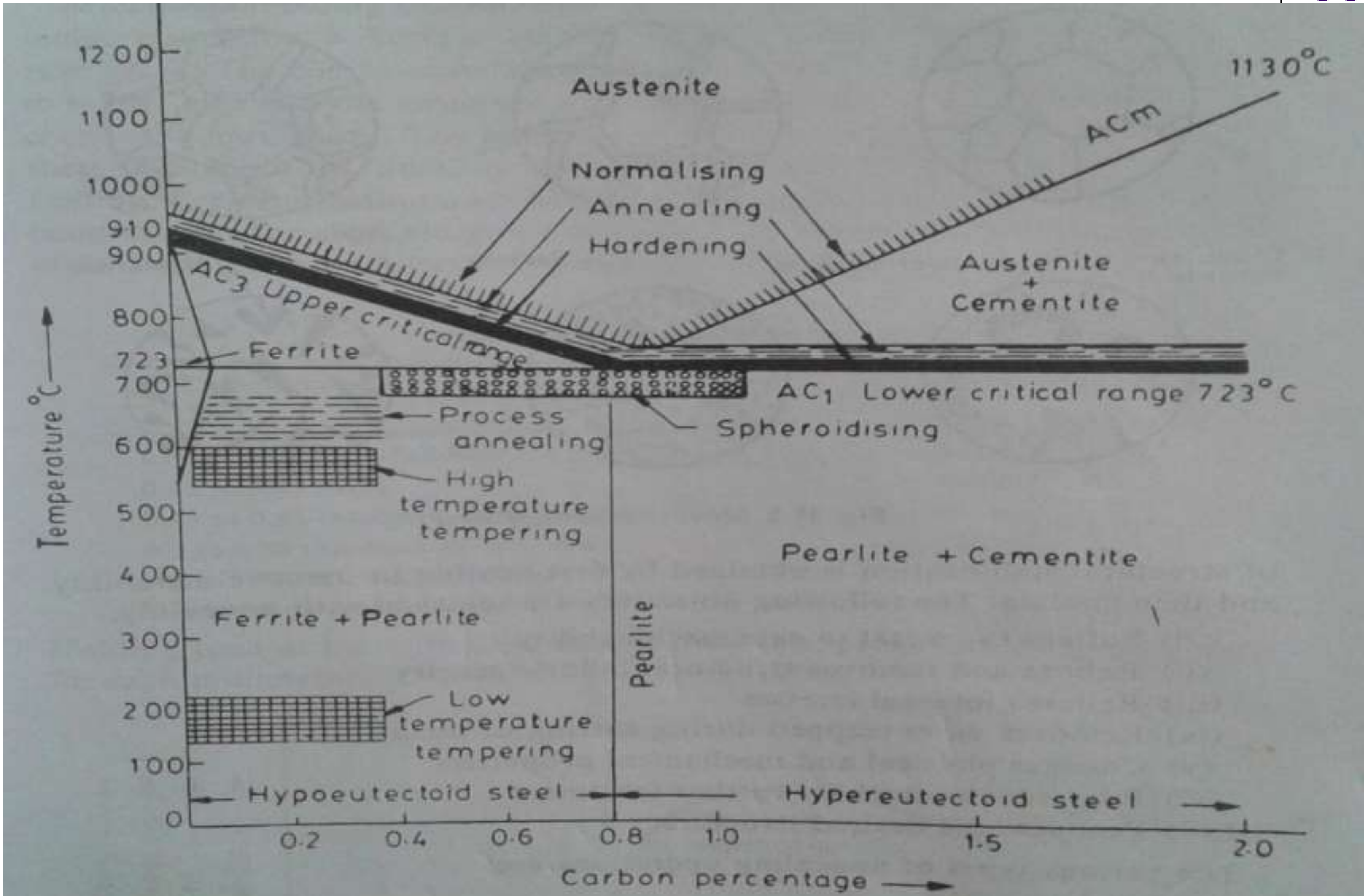
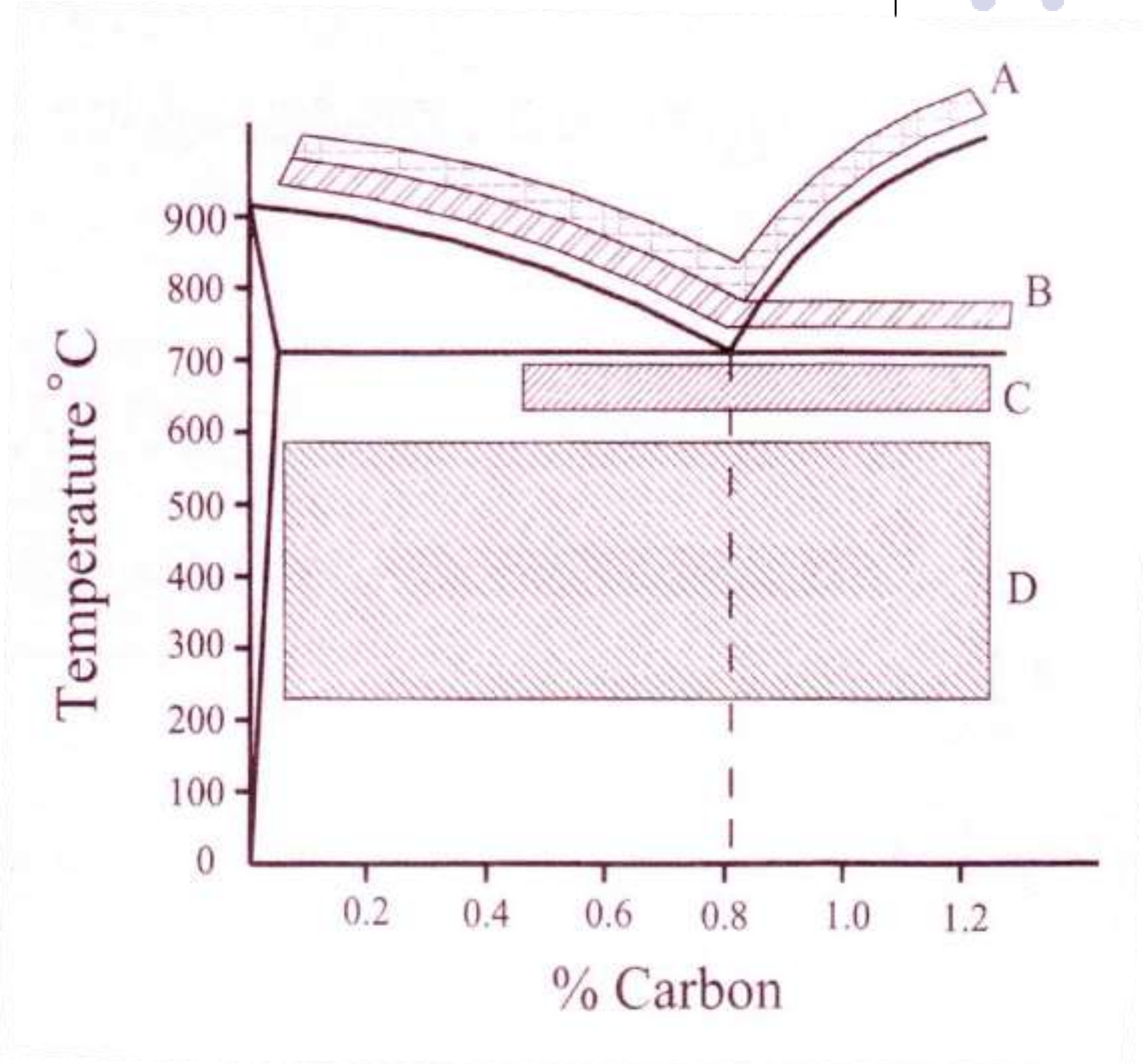


Fig. 11.1 *Heat treatment ranges and processes*

Heat Treatments

- A – Normalising
- B – Annealing or Hardening
- C – Spheroidising or Process Annealing
- D - Tempering



Annealing

- Any heating and cooling operation that is usually applied to induce softening
- Depending on the specific purpose , annealing process are classified as:
- Full annealing
- Process annealing
- Spheroidise annealing

Full annealing

Purposes

- To soften the steel
- To relieve internal stresses
- To refine grains

Process

- Heating the steel 30 -50 degree above UCT for hypo-eutectoid steel and by same temp. above LCT for hyper-eutectoid steel
- It is held at this temp. for some time and then cooled slowly

Normalising

Purposes

- To refine the grain structure
- To relieve internal stresses
- To improve machinability, tensile strength etc

Process

- Heating the steel 30 – 50 degree above its UCT for hypo and hyper eutectoid steels, holding at this temp. for shorter time and then allowed to cool down in still air.

Hardening

Purposes

- _to increase the hardness of metal so that it can resist wear
- To enable it to cut other metals

Process

- Heating the steel 30 -50 degree above UCT for hypo-eutectoid steel and by same temp. above LCT for hyper-eutectoid steel
- It is held at this temp. for considerable time and then quenched in a suitable cooling medium

Tempering

Purposes

- To reduce the brittleness of hardened steel.
- To increase toughness and ductility
- To remove internal stresses
- *Process*
- Reheating the hardened steel to some temperature below LCT , followed by any desired rate of cooling.

Surface hardening/Case hardening

- In many important industrial products should have a hard, wear resistant surface and soft and tough core

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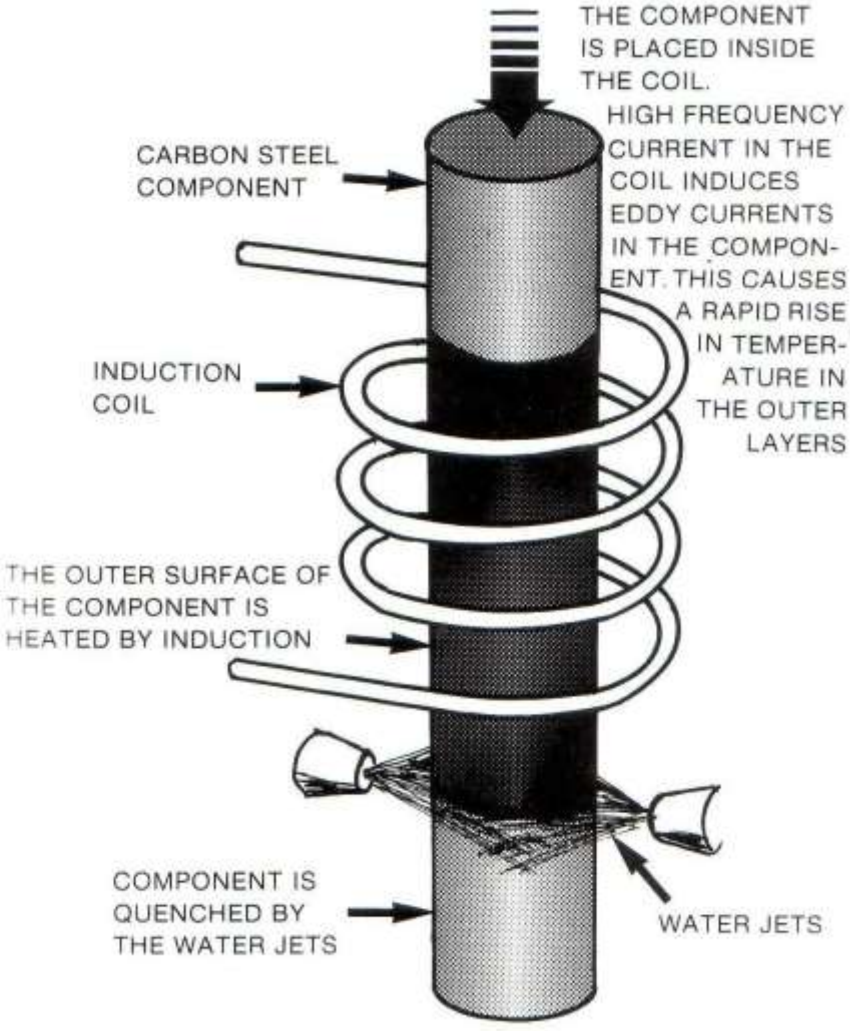
- Case hardening seeks to give a hard outer skin over a softer core on the metal.
- The surface layers of steel plates heated to the hardening temp. And it is then quenched in water or some other medium.

SURFACE HARDENING PROCESSES

- Carburising – introducing *carbon* in to the surface of steel. In this process low carbon steel is heated in contact with carbonaceous material from which steel absorbs carbon
- Nitriding – introducing *nitrogen* in to the surface of steels by heating and holding it at a suitable temp. In contact with paritially dissociated ammonia or suitable medium
- Cyaniding – both *carbon and nitrogen* are added to the surface layers of steel

Induction hardening

- Induced eddy currents heat the surface of the steel very quickly and is quickly followed by jets of water to quench the component.
- A hard outer layer is created with a soft core.
- The slideways on a lathe are induction hardened.



Flame hardening

- Gas flames raise the temperature of the outer surface above the upper critical temp.
- Water jets quench the component and thereby hardens it .